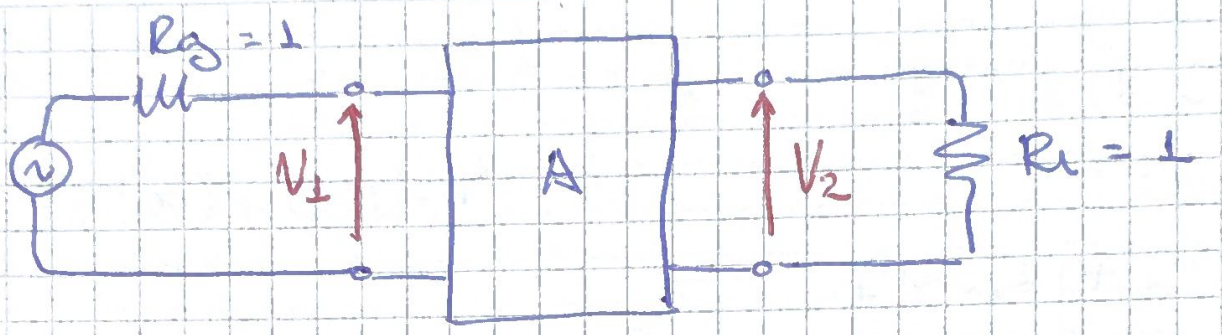
TS10 - TC2

Lucas Gallero.

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31-10-26



Requirimientos

- LPF - Cheby. - orden 3
- No Disipativo → LC
- Normalizado en frec. e impedancia

$$\text{Cheby} \rightarrow |T|^2 = \frac{1}{1+\varepsilon^2 C_n^2}$$

$C_0 = 1$

$C_1 = w$

$C_2 = (2w \cdot w) - 1 = 2w^2 - 1$

$C_3 = (2w \cdot (2w^2-1)) - w) = 4w^3 - 2w - w = 4w^3 - 3w$

$$|T|^2 = \frac{1}{1+\varepsilon^2 (4w^3-3w)^2}$$

Adopto $\alpha_r = 0{,}5$ dB

$\varepsilon = 0{,}3493$

$$\varepsilon = \sqrt{10^{\frac{0{,}5}{10}} - 1} = 0{,}3493 \quad \Rightarrow \quad \varepsilon^2 = 0{,}122$$

$$a = \frac{1}{n} \sinh^{-1}\left(\frac{1}{\varepsilon}\right) \qquad \theta_k = \frac{(2k-1)\pi}{2n} \rightarrow \theta_1 = \pi/6$$

$\theta_2 = \pi/2$

$\theta_3 = \frac{5}{6}\pi$

$$\begin{cases} -\sigma_k = \sin(\theta_k)\sinh(a) \\ \pm w_k = \cos(\theta_k)\cosh(a) \end{cases}$$

Papel de fibra de caña de azúcar.

$a = 0{,}5914$

$p_1 = -0{,}3132 + j\,1{,}0219 \longrightarrow$
$p_2 = -0{,}6265 \downarrow$
$p_3 = -0{,}3132 - j\,1{,}0219 \longrightarrow$

$\Big\} \therefore s^2 + 2\alpha s + (\alpha^2 + \beta^2)$

$(s + 0{,}6265)(s^2 + 0{,}6264\,s + 1{,}1424)$

$s^3 + 1{,}2529\,s^2 + 1{,}5349\,s + 0{,}7157.$

$$T(s) = \frac{0{,}7157}{s^3 + 1{,}2529\,s^2 + 1{,}5349\,s + 0{,}7157}$$

$$|S_{21} = T(s) =|$$

$$|S_{21}|^2 = |T(j\omega)|^2 = \frac{1}{1 + \varepsilon^2(4\omega^3 - 3\omega)^2} = \frac{1}{1 + A}$$

$A = \varepsilon^2(4\omega^3 - 3\omega)^2$

$|S_{21}|^2 + |S_{11}|^2 = 1 \quad \therefore \quad |S_{11}|^2 = 1 - |S_{21}|^2$

$$|S_{11}|^2 = 1 - \frac{1}{1+A} = \frac{1 + A - 1}{1 + A} = \frac{A}{1+A}$$

$$|S_{11}|^2 = \frac{\varepsilon^2(4\omega^3 - 3\omega)^2}{1 + \varepsilon^2(4\omega^3 - 3\omega)^2} = |\varepsilon(4\omega^3 - 3\omega)|^2 \cdot |S_{21}|^2$$

$$|S_{11}| = \varepsilon |4w^3 - 3w| \cdot |S_{21}| \qquad j^3 = -j \quad j^2 = -1$$

$$s = jw \rightarrow w = -js$$

$$4w^3 - 3w = 4(-js)^3 - 3(-js) = 4js^3 + 3js$$

$$4w^3 - 3w = j(4s^3 + 3s)$$

$$|4w^3 - 3w| = |4s^3 + 3s|$$

$$\boxed{S_{11}(s) = \varepsilon(4s^3 + 3s)\, S_{21}(s)}$$

$$k = \frac{1}{2^{n-1}\varepsilon} = \frac{1}{2^{3-1}\varepsilon} = \frac{1}{4\varepsilon} = \frac{1}{4 \cdot 0{,}3493} = 0{,}7157$$

$$S_{11} = \varepsilon(4s^3 + 3s) \cdot k \cdot \frac{1}{D(s)}$$

$$S_{11} = \cancel{\varepsilon}(4s^3 + 3s) \cdot \frac{1}{4(\cancel{\varepsilon})} \cdot \frac{1}{D(s)}$$

$$\boxed{S_{11} = \frac{s^3 + \frac{3}{4}s}{s^3 + 1{,}2529\,s^2 + 1{,}5349\,s + 0{,}7157}} = \frac{F}{G}$$

$$Z_1 = \frac{F+G}{F-G} = \frac{2s^3 + 1{,}2529\,s^2 + 2{,}2849\,s + 0{,}7157}{\cancel{(}1{,}2529\,s^2 + 0{,}1784\,9s + 0{,}7157\cancel{)}^{(-1)}}$$

Papel de fibra de caña de azúcar.

$$Z_1 = \frac{2S^3 + 1{,}252913\,S^2 + 2{,}2849\,S + 0{,}7156\,94}{1{,}2\,\Leftarrow 2913\,S^2 + 0{,}784895\,S + 0{,}7156\,94.}$$

$2S^3 + 1{,}252913S^2 + 2{,}2849\,S + 0{,}715694 \;\big|\; D_{Z1}(S)$

$-\;2S^3 + 1{,}252913S^2 + 1{,}14248\,S$

$1{,}59628\,S$

$0 + 0 + 1{,}1424\,S + 0{,}715694$

$L_1 = 1{,}59628\,H$

$1{,}252913S^2 + 0{,}784895\,S + 0{,}715694 \;\big|\; 1{,}1424\,S + 0{,}715694$

$1{,}09669\,S$

$1{,}252913S^2 + 0{,}7848959\,\lambda$

$0 \qquad 0 + 0{,}715694$

$1{,}09669\,F$

$1{,}1424\,S + 0{,}715694 \;\big|\; 0{,}715694$

$1{,}1424\,S \qquad\qquad 1{,}59628\,S$

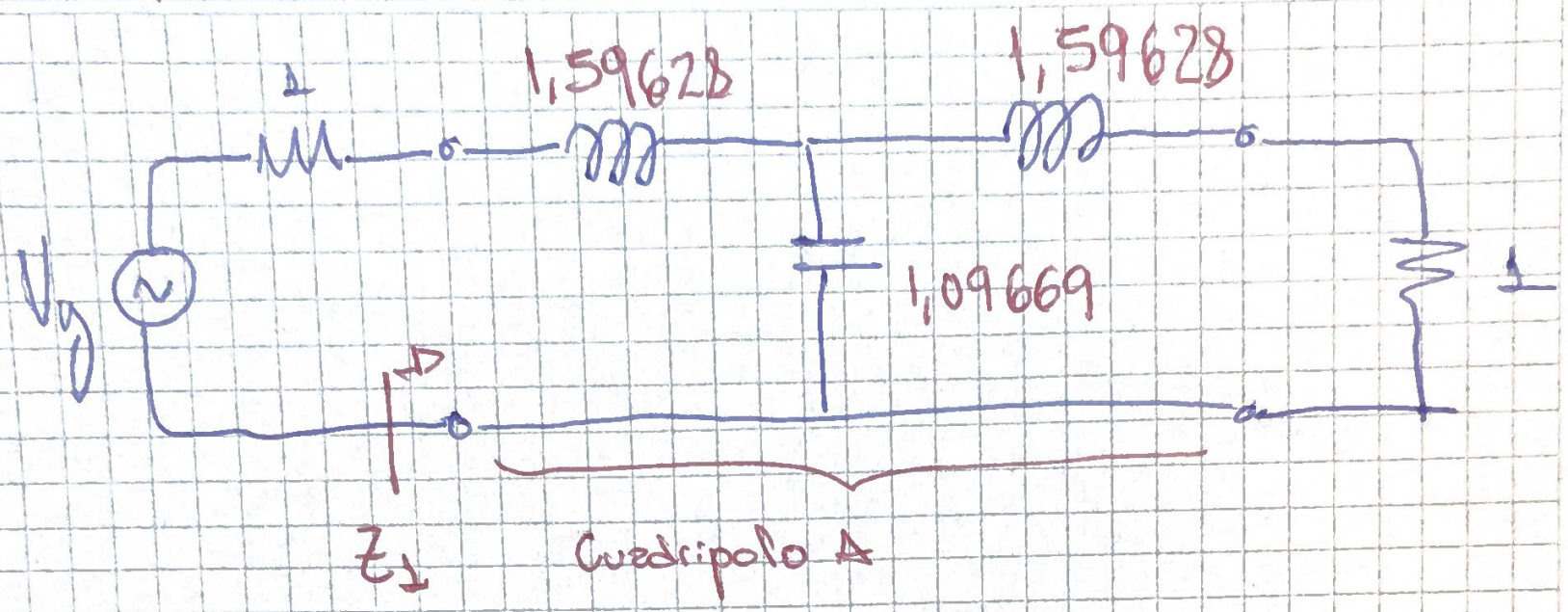
$L_2 = 1{,}5928\,H$

$0{,}715694$

$0{,}715694 \;\big|\; 0{,}715694$

$0/ \qquad 1/$

$1\,F$

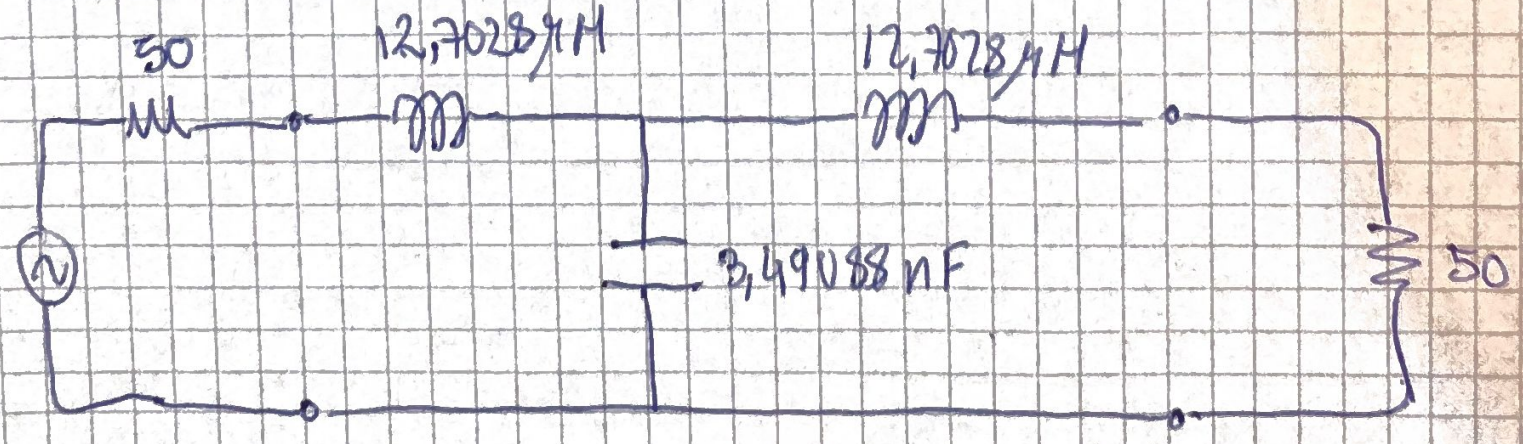


(5) $\omega_c = 2\pi \cdot 10^6$ rad/seg $\qquad R_g = 50\Omega \rightsquigarrow R_0 = 50\Omega$

$f_c = \frac{\omega_0}{2\pi} = 1\,MHz$

Desnormalización

$$\begin{cases} R = R_0 R_n \rightarrow R_g = R_L = 50\Omega \\ L = \frac{R_0 L_n}{\omega_c} \rightarrow L_1 = L_2 = \frac{50 \cdot 1{,}59628}{2\pi \cdot 10^6} = 12{,}7028\,\mu H \\ C = \frac{C_n}{R_0 \omega_c} \rightarrow C = \frac{1{,}09669}{50 \cdot 2\pi \cdot 10^6} = 3{,}49088\,nF \end{cases}$$



Papel de fibra de caña de azúcar.